Lecture 1: Section 0.1

Function

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What is a Function?

Definition

A **function** is a relation between sets where each input (x-value) is associated with **exactly one** output (y-value).

Example

Which of these represent functions?

- $f(x) = x^2 + 3x - 2$
- $x^2 + y^2 = 25$
- $y = \sqrt{x - 4}$
- $\{(1, 2), (3, 5), (1, 4), (7, 8)\}$

Function Notation

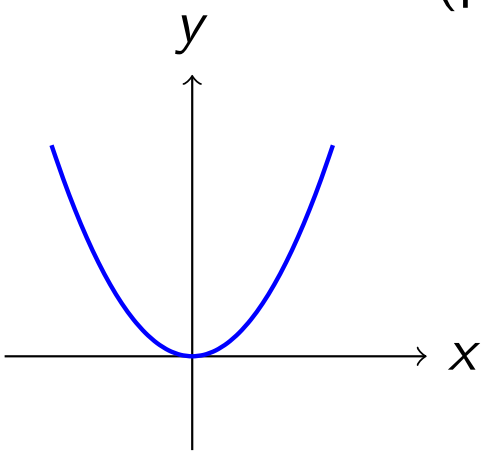
$f(x)$ is read as "f of x" and represents the output value when x is the input.

Vertical Line Test

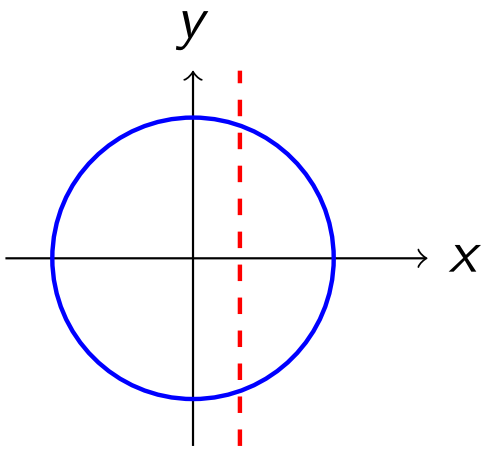
The Test

If any vertical line intersects a graph at **more than one point**, the graph does NOT represent a function.

Function (passes test)



Not a function (fails test)



Practice

Example

Use the vertical line test to determine which graphs represent functions:

- A straight line
- A circle
- A parabola opening upward
- A sine curve

Piecewise Functions

Definition

A **piecewise function** is defined by different formulas for different parts of its domain.

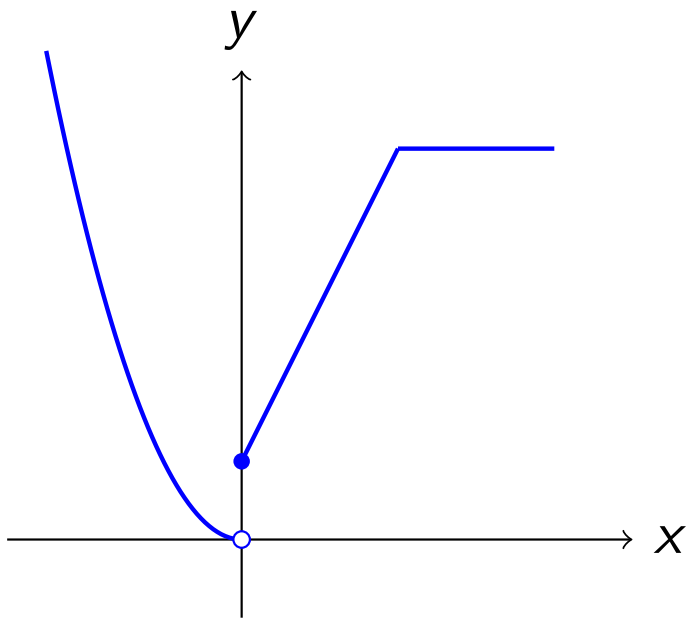
Example

Graph and evaluate the piecewise function:

$$f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 2x + 1 & \text{if } 0 \leq x \leq 2 \\ 5 & \text{if } x > 2 \end{cases}$$

Find: $f(-2)$, $f(0)$, $f(1)$, $f(3)$

Piecewise Functions



Domain and Range

Domain

The set of all possible **input values** (x-values) for which the function is defined.

Range

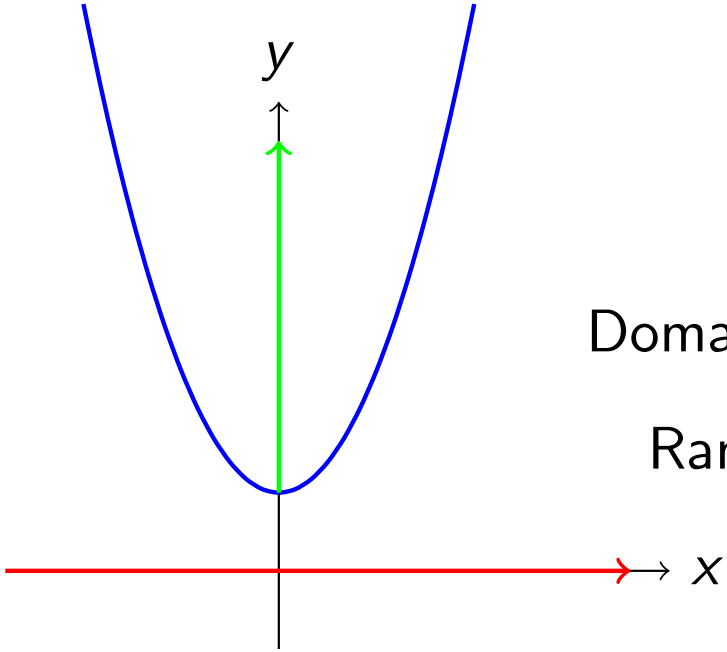
The set of all possible **output values** (y-values) that the function can produce.

Example

Find the domain and range of:

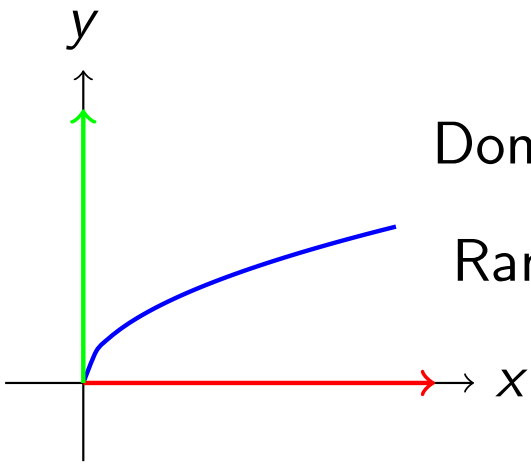
- $f(x) = x^2 + 3$
- $g(x) = \sqrt{x - 4}$
- $h(x) = \frac{1}{x-2}$
- $k(x) = |x + 1|$

Domain and Range Visualization



Domain: $(-\infty, \infty)$

Range: $[3, \infty)$



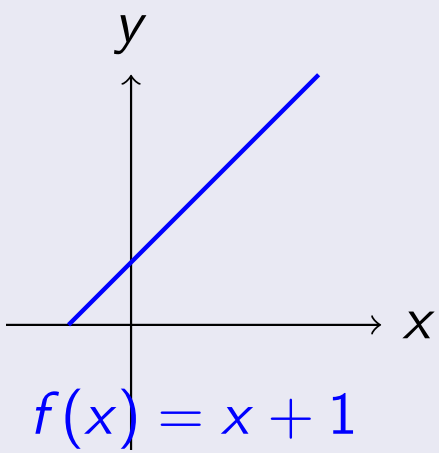
Domain: $[0, \infty)$

Range: $[0, \infty)$

Common Function Graphs (Part 1)

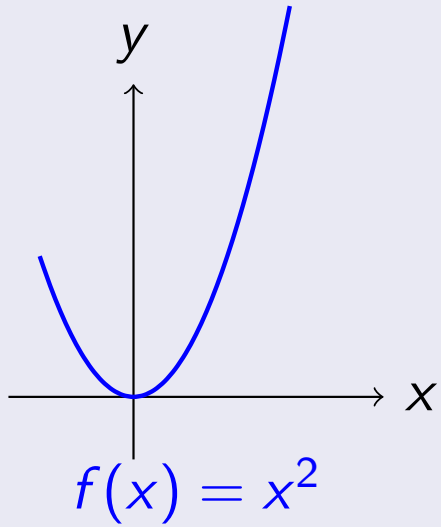
Linear Function

$f(x) = mx + b$



Quadratic Function

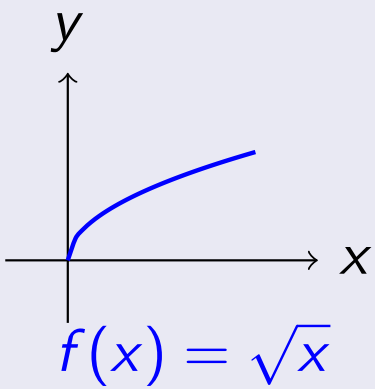
$f(x) = ax^2 + bx + c$



Common Function Graphs (Part 2)

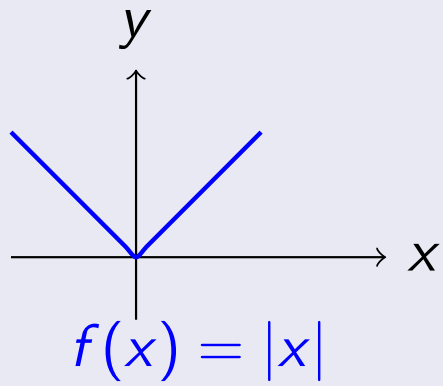
Square Root Function

$f(x) = \sqrt{x}$



Absolute Value Function

$f(x) = |x|$



Practice Problems

Function Identification

1. Which of the following relations represent functions?
   - $\{(1,2),(3,4),(5,6),(7,8)\}$
   - $\{(1,2),(1,3),(2,4),(3,5)\}$
   - $y^2 = x$
   - $y = 2x + 3$

Piecewise Functions

1. Graph and evaluate:

$$f(x) = \begin{cases} x+2 & \text{if } x < 1 \\ x^2 & \text{if } x \geq 1 \end{cases}$$

Find $f(0)$, $f(1)$, $f(2)$

Practice Problems

Domain and Range

1. Find domain and range of: $f(x) = \sqrt{4 - x^2}$
2. Find domain of: $g(x) = \frac{1}{x^2 - 9}$

Applications of Functions

Real-world Applications

- Physics: Position, velocity, and acceleration as functions of time
- Economics: Cost, revenue, and profit functions
- Engineering: Signal processing and control systems
- Computer Science: Algorithms and data structures

Key Concepts

- A function assigns exactly one output to each input
- Use vertical line test to identify functions graphically
- Piecewise functions use different rules for different inputs
- Domain: what you can put into the function
- Range: what you get out of the function